LESSON PLAN

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Name: Jett Dobson **Cohort: EDUC 4355 - Effective Intermediate Teaching**

Lesson Plan

Lesson Title: Conductors vs Insulators: Do Charges Move or Stay?	**Date:** February 13, 2026
Grade Level: 9 (SNC1W)	**Subject/Strand:** Science (destreamed), Strand D (Physics)
Topic: Conductivity (holding vs transferring electric charge)	**Length of Period:** 70 Minutes
Lesson Plan Description: Students will investigate a variety of everyday materials to determine whether they transfer electric charge (conductors) or hold electric charge (insulators), and use evidence from their tests to justify their classifications.	

STEP 1: CURRICULUM CONNECTIONS

Ontario Curriculum Overall Expectations	**Ontario Curriculum Specific Expectations**
D2. Investigating and Understanding Concepts Demonstrate an understanding of the nature of electric charges, including properties of static and current electricity	**D2.2** Determine the conductivity of various materials by investigating their ability to hold or transfer electric charges
Learning Goals	**Success Criteria**
We are learning to test different materials and use evidence to decide if they let electric charge move through them (conductors) or block charge (insulators)	I can … 1. Build and use the conductivity tester safely and correctly 2. Test materials fairly and repeat a trial if my result is unclear 3. I can explain, using my test results, why a material is a conductor or an insulator, using the correct science terms (electric charge, conductor, insulator). 4. Collaborate with my group by doing my role, listening, and helping my group stay on task.

STEP 2: ASSESSMENT

Indicate purpose of the assessment : **[X] *FOR*** **[X] *AS*** **[ ] *OF***

Indicate Achievement Chart categories being assessed
[ ] *Knowledge and Understanding* **[X] *Thinking*** **[ ] *Application*** **[X] *Communication***

Indicate Learning Skills/Work Habits:
[ ] responsibility, [ ] organization, [ ] independent work, [X] collaboration, [ ] initiative, [ ] self-regulation
(Incorporate one with Learning Goal and Success Criteria)

Indicate Assessment Mode: *Written, Oral, Performance* (Write, Say, Do)	**Indicate Assessment Strategy :** What will learners do to demonstrate their learning?	**Indicate Assessment Tool:** Instrument used to record results/ document learning.
Written (*Write*): Structured note sheet (predictions+data tables), evidence statements, 3-2-1 exit ticket. **Oral (*Say*):** Think-pair-share responses, midpoint checkpoint conference explanation, group share for the T-chart. **Performance (*Do*):** Build and use the conductivity tester safely, conduct fair trials and repeat tests for reliability, complete the static "hold charge" test.	● Record investigation results on the structured note sheet (transfer and static tests). ● Justify classifications with brief evidence statements (conductor/insulator). ● Share one evidence-based result during the checkpoint/T-chart. ● Complete the 3-2-1 exit ticket (learned, interesting, question).	● Structured note sheet (predictions + data tables + evidence statements) (AFL) ● Teacher observation checklist during investigation and group work (AFL) ● Midpoint checkpoint conference notes (AFL) ● 3-2-1 exit ticket (reflection + remaining question) (AFL + AAL)

STEP 3: CONSIDERATIONS FOR PLANNING

Prior Learning: What prior experiences, knowledge and skills do the learners bring with them to this learning experience?
Students may bring with them:

- Everyday experiences with static electricity (doorknob shocks, balloon on hair, clothes clinging) and the ability to describe observations.
- Prior unit learning that objects can become charged through friction/contact, that opposite charges attract and like charges repel, and that static electricity involves charge build-up and possible discharge.
- Practical understanding that electrical devices require a complete path to work (flashlights, chargers, switches), even if circuits have not been formally modelled.
- Familiarity with common materials (metals, plastics, rubber, wood, paper) and informal rules (metal conducts; plastic insulates), plus common misconceptions (confusing with magnetism; assuming shiny/heavy means conductive).
- Experience with investigation routines (short procedures, materials-bin use, group roles, recording in tables) and emerging data habits (using categories such as bright/dim/off and none/some/lots; repeating unclear trials).
- Group communication skills, including sharing observations, listening to peers, and beginning to separate observation from interpretation.

IEP program implications: Accommodations, Modifications -- To be completed for Placements 2, 3 and 4

1. **IEP: Learning disability - communication**

Instructional support in this lesson: pre-teach and re-teach key terms (conductor, insulator, transfer, hold, evidence, reliable) using the visual word bank; provide chunked directions with a "repeat-back" of the next step before starting each phase.
Organization supports: single-page structured note with pre-made tables, checkboxes, and sentence frames; teacher prompts to record observations first (bright/dim/off; none/some/lots) before interpreting.
Attention supports: neutral attention cue/prompt to return to task; built-in "gear shifts" (predict → build → test 4 → checkpoint → test 4 → static → evidence)
Environment/grouping: strategic seating within the group, placed with mature/supportive peers; clearly defined role to reduce social friction and increase participation.
Assessment access: allow speech-to-text or typing for evidence statements; permit brief oral check-in at the midpoint checkpoint to demonstrate reasoning if writing is a barrier; provide additional time for the final evidence statements/exit ticket if needed; quiet/low-distraction option for the reflection portion.

2. **Non-identified IEP: Assistive Technology (Accommodations)**

Instructional supports in this lesson: provide directions in short numbered steps with the tester diagram; read key instructions aloud once and keep them visible at the station.
Assistive tech access: digital version of the structured note when helpful; text-to-speech for reading prompts; speech-to-text or word processing for evidence statements; spelling/grammar supports enabled as needed.
Writing-load reduction: allow labelled diagrams and brief point-form notes where appropriate, while still requiring an evidence-based classification statement.
Self-advocacy support: establish a discreet help signal and schedule a quick check-in during the midpoint checkpoint so support does not rely on the student initiating it.
Assessment focus: emphasize accuracy of observations and evidence-based reasoning rather than writing length/mechanics; allow oral explanation during checkpoint to confirm understanding.

3. **IEP: Behavioural (ADHD/anxiety; self-regulation focus)**

Structure and transitions: agenda and timings posted; 2-minute transition warnings; stable classroom layout and predictable routines throughout the investigation.
Task management: highly structured activity sequence with a visible "finish line" checklist (build → test 4 → checkpoint → test 4 → static → evidence); chunk work into short parts and use prompts to return to task.
Environment: strategic seating away from high-traffic areas; structured workspace with only required materials out at each phase to reduce overstimulation.
Regulation supports: planned periodic breaks; option for a brief reset during the checkpoint if needed; quiet/individual setting available for the final writing portion if dysregulated. Can use a break signal card to notify personnel whenever they may need a break.
Assessment access: word processing/digital organizer option for recording; extra processing time or spaced completion of reflection if needed; evidence gathered through the organizer plus brief oral check-in at checkpoint to demonstrate understanding even if writing output is reduced in the moment.

Differentiation -- How will I differentiate the instruction to ensure the inclusion of all learners? (Choose 1 or 2 areas in Year 1)

Content *(what learners are going to learn, and when):* Students learn the difference between materials that transfer charge (conductors) and block/hold charge (insulators/static effects) through a brief review and demo in Minds On, then apply the concept during two linked investigations in Action. Vocabulary and key ideas are supported throughout with a visible word bank and a single, consistent focus on evidence-based classification.
Process *(how instruction is organized towards a gradual release of responsibility):* Instruction follows gradual release: teacher models the tester and one data row, groups practise with structured roles and a common testing sequence, then students independently justify

classifications using evidence statements. A midpoint checkpoint conference provides timely feedback and prompts students to improve reliability before completing the second half of testing.

Product *(the ways in which individual learners will demonstrate learning for assessment)*: Students demonstrate learning by completing the structured note sheet (data tables, checkpoint reflection, evidence statements) and the flexible 3-2-1 exit ticket, with options to respond in point form, typed, or orally during teacher check-ins. Oral evidence sharing during the checkpoint/T-chart provides an additional way to communicate reasoning without relying only on writing.

Environment *(the context and environment in which learners learn and demonstrate learning):* Learning occurs in predictable, organized lab-style stations with labelled bins, posted procedures, and a projected timer to support attention and smooth transitions. A low-distraction option and planned brief resets are available, and assistive technology supports are built in so students can access and record learning in ways that reduce barriers.

Resources and Materials & Technology Integration *Hand outs / other resources are not required for submission of this lesson plan.*

Resources, Materials and Technology integration for class of 25 (22 + 3)

- PowerPoint slide deck (group list/table numbers, "Static or Current?" warm-up, learning goal, success criteria, fair-test reminders, agenda/timing) + projected timer.
- Conductivity tester diagram, fair-test rules, and visual word bank (projected and/or 1 printed copy per group).
- Student handout: Conductivity Investigation Notes (25 copies; includes predictions, data tables, midpoint checkpoint, evidence statements, 3-2-1 exit ticket) + digital fillable version for assistive technology (speech-to-text/text-to-speech enabled).
- Teacher assessment tool: observation checklist/anecdotal notes (1).

Group materials (7 groups: 4 groups of 4; 3 groups of 3):

- Labeled bin/tray with low-voltage power source, LED (and resistor if needed), 3–4 alligator leads, and the 8 test materials (coin, paperclip, foil, graphite pencil lead, rubber band, plastic spoon, wood stick, paper).
- Static kit: balloon or plastic comb + paper bits.

Teacher materials:

- Pre-built conductivity tester for modelling/troubleshooting; known conductor/insulator for checks (paperclip/plastic); spare batteries/LEDs/leads.

Three Part Lesson

Indicate Instructional Strategy

Modelled → Guided → Small group investigation → Independent exit ticket (teacher conferencing throughout).

Minds on: Motivational Hook/engagement /Introduction (approximately 10-20%)

Estimated time (10-12 minutes)

1. Greet students at the door and direct them to pre-assigned investigation groups (group list and table numbers on the opening slide). Remind students the materials bins remain closed until instructed.
2. Display the "Static or Current?" slide. Read three everyday scenarios aloud; after each, students show 1 finger (static) or 2 fingers (current). Confirm the correct choice in one sentence and move to the next scenario.
3. Run a 30-second think-pair-share: "From last class, what has to happen for an object to become charged?" Circulate, take two quick shares, and restate the key ideas (charge can build up; attraction/repulsion; discharge can occur).
4. Show the conductivity tester students will use. Complete two observation-only trials (paperclip: LED on; plastic spoon: LED off). Do not explain the reason yet; remind students they will use evidence to identify patterns during the investigation.
5. Display and read the Learning Goal, then display and read the Success Criteria. Have students locate these on their notes so they are anchored before the investigation begins.
6. Before individual writing, prompt groups: "Choose one material you predict will turn the LED on and one you predict will not, and be ready to give one short reason." Cold-call two groups for one sentence each.
7. Students complete the prediction section for the eight test materials and write one reason for two predictions using the provided sentence

Prompting Questions and Possible Learner Responses:

Q1.Static or current: balloon on hair, phone charging, doorknob shock

A1. Balloon on hair and doorknob shock are static (charge builds up; shock is discharged). Phone charging is current (charge moves through a complete path).

Q2. From last class, what has to happen for an object to become charged?

A2. Charge must transfer between objects (often by friction or contact), leaving one with more electrons and the other with fewer.

Q3. What happens when opposite charges are near each other? What about like charges?

A3. Opposite charges attract; like charges repel.

Q4. In the demo, what did you observe with the paperclip vs the plastic spoon?

A4. Paperclip: LED on. Plastic spoon: LED off.

Q5. Why might the LED turn on for one material but not the other?

A5. Conductors allow charge to move through; insulators block charge movement.

Q6. What counts as evidence today?

A6. LED result (bright/dim/off), repeatability, consistent

stem. Circulate and give quick prompts to support students who need help starting.

8. Close with a readiness check: "What will count as evidence today?" Accept 2–3 responses (LED bright/dim/off; repeating a trial; consistent contact method), then state the transition to the Action: students will test materials and use evidence to classify them.

Considerations

Keep instructions short, visible, and read aloud; maintain frequent gear shifts; accept oral responses. Keep sentence stems/word bank visible; provide digital notes and speech-to-text for assistive tech; use predictable check-ins and clear routines for behavioural needs; chunk directions and prompt "What's your next step?" for learning disability support.

method; static attraction (none/some/lots).

Q7. What makes the test fair and results reliable?

A7. Same circuit and battery, consistent contact points/method, repeat unclear trials, and check the tester with a known conductor if results seem off.

Can use these questions flexibly as a question bank to check understanding, extend thinking, and support groups as needed.

Action: During /Working on it (approximately 60%)

Estimated time: 45 minutes

Transition from Minds On: Students move from predictions to evidence collection. They will investigate which materials **transfer electric charge** by completing a circuit (conductivity test) and which materials appear to **hold charge** through static attraction (static test). Students record observations first, then classify materials using evidence.

Materials and setup: Each group has a labelled bin containing a low-voltage power source (battery pack), LED (and resistor if required), 3–4 alligator leads, a testing set of eight materials (coin, paperclip, foil, graphite pencil lead, rubber band, plastic spoon, wood stick, paper), a static kit (balloon or plastic comb and paper bits), and one tester diagram/fair-test reminder. Each student has a structured note sheet for predictions, data tables, evidence statements, and the midpoint checkpoint.

1. **Role assignment and expectations (~2 minutes)**

Groups confirm roles (Builder, Tester, Recorder, Materials Manager; groups of three omit Materials Manager). Expectations for safe handling and fair testing are reviewed briefly: consistent setup, consistent contact method, repeat trials when unclear, and record observations before interpretation.

2. **Teacher modelling of procedure and recording (~4 minutes)**

The teacher models one complete test trial and demonstrates how to record results using the class categories (LED bright/dim/off; repeat check yes/no; classification conductor/insulator/uncertain). The teacher also models how to respond to unclear results (improve contact, repeat the trial, verify with a known conductor).

3. **Group construction of conductivity tester (~6 minutes)**

Groups assemble the tester following the diagram. The teacher circulates to check each group's setup before testing begins (LED orientation, secure connections, visible test gap, and confirmation that the circuit functions using a known conductor).

4. **Conductivity investigation: transfer charge (Part 1) (~10 minutes)**

Groups test the first four materials in a common order (coin, paperclip, foil, graphite) to maintain pacing and comparability. The Recorder documents observations and repeat checks, then classifies each material using evidence. The teacher circulates and provides descriptive feedback focused on fair testing and evidence language.

5. **Midpoint checkpoint and role rotation (~6 minutes)**

Groups pause after four materials. Students complete the checkpoint on their note sheet (one fairness action, one possible source of error, and one result they trust with justification). The teacher conducts brief conferences with each group to confirm reliability and provide one targeted improvement. Roles rotate once to ensure shared participation.

6. **Conductivity investigation: transfer charge (Part 2) (~8 minutes)**

Groups test the remaining four materials (rubber band, plastic spoon, wood stick,

Prompting Questions and Possible Responses:

Critical thinking questions used throughout the lesson

Q1. What observation are you recording before you decide what it means?

A1. Whether the LED is bright, dim, or off (and whether it repeats the same way).

Q2. What does it mean if the LED lights consistently when the material bridges the gap?

A2. The material allows charge to transfer through it, so it is acting as a conductor.

Q3. What could cause a dim or inconsistent result besides the material itself?

A3. Poor contact with the clips, loose wires, low battery power, LED orientation, or the material not touching the metal clips well.

Q4. How are you keeping the test fair across materials?

A4. Using the same circuit and battery, using the same contact method and points, testing in the same order, and repeating unclear trials.

Q5. Why is repeating a trial important in this investigation?

A5. It helps confirm the result is reliable and not caused by a one-time error.

Q6. Which row in your table do you trust most, and why?

A6. A row where the LED result was clear and repeated the same way, and the contact method was consistent.

Q7. What pattern do you notice about materials that conduct well?

A7. Metals tend to make the LED bright; non-metals like plastic and rubber tend to keep it off.

Q8. In the static test, what does "lots of attraction" suggest?

A8. The material shows evidence of charge build-up/holding that creates attraction to the paper bits.

Q9. How is "holding charge" in the static test different from "transferring charge" in the circuit test?

A9. Holding charge is about charge building up on a surface and causing attraction, while transferring charge is about charge moving through a continuous path.

Q10. What evidence will you use in your final classification statement?

paper) and complete the table using the same recording conventions and repeat-check expectation for unclear results.

7. **Static investigation: hold charge (~7 minutes)**

Groups complete a short, consistent static test using the balloon/comb and paper bits. Students test the same three materials (plastic spoon, rubber band, paper) and record attraction as none/some/lots, repeating once for reliability. Students briefly note how this "holding charge" evidence differs from the transfer test.

8. **Evidence statements (~4 minutes)**

All students complete two evidence-based statements on their note sheet: one classification (conductor/insulator) justified by circuit observations and one statement connecting the static observation to the idea of charge holding. The teacher circulates to ensure every student produces evidence-based reasoning (written, typed, or orally recorded as per accommodations).

IEP Supports

Use visible timing and predictable phases; provide chunked directions and repeat-back; use the single-page organizer with tables and sentence frames; allow digital notes, speech-to-text, and text-to-speech; use checkpoint conferences to provide an oral pathway for demonstrating reasoning when writing is a barrier.

A10. The LED result (bright/dim/off), whether it repeated consistently, and the static attraction result (none/some/lots) if relevant.

Can use these questions flexibly as a question bank to check understanding, extend thinking, and support groups as needed.

Consolidation & Connection (Reflect and Connect) (approximately 10 - 20%)

Estimated time: 10- 13 minutes

1. **Quick class check-in and T-chart (~6 minutes)**

The teacher begins with a quick whole-class synthesis using a T-chart that is already prepared on the board with the headings **Conductors (transfer charge)** and **Insulators (do not transfer easily)**. Each group shares one tested material and one observation using the sentence frame, "We tested _____ and the LED was _____, so we classified it as _____," and the teacher records the example on the chart.

2. **3-2-1 Exit Ticket (~6 minutes)**

The teacher keeps the word bank visible (conductor, insulator, transfer, hold, reliable; bright, dim, off) and models one example learning statement. Students complete a flexible 3-2-1: 3 things I learned today (any three learning statements), 2 things I found interesting (two observations/ideas), and 1 thing I still have questions about (one remaining question). Responses may be full sentences or point form. The teacher collects exit tickets at the door to inform next-lesson grouping and support.

Student and IEP considerations (exit ticket)

Word bank remains visible; students may respond in point form. Learning disability: use sentence frames; teacher check-in/scribing if needed; complete in order (3, then 2, then 1). Assistive technology: digital or speech-to-text option; reduce writing load and prioritize meaning. Behavioural: clear timer/finish line, brief reset option, quiet seating and predictable teacher check-in.

Prompting Questions:

Questions used to assess learners' understanding

Q1. Name one material you classified as a conductor or insulator and state your evidence.

A1. "Paperclip is a conductor because the LED was on/bright." or "A plastic spoon is an insulator because the LED stays off."

Q2. What does the LED turning on tell you about charge in that material?

A2. "Charge can transfer through it and complete the circuit."

Q3. How did you make your results more reliable when something was unclear?

A3. "We repeated the trial, improved contact, kept contact points consistent, and checked with a known conductor."

Q4. How is transferring charge in the circuit test different from holding charge in the static test?

A4. "Transferring is charge moving through a path; holding is charge building up on a surface and causing attraction."

Q5. Why do real devices need both conductors and insulators?

A5. "Conductors carry electricity where it needs to go; insulators prevent shocks and short circuits."

Extension Activities -- What will learners do when work is completed? What will learners do if they finish early?

1. Test a "mystery" teacher-approved item and justify conductor/insulator using one evidence sentence.
2. Re-test graphite or another "dim" material and explain two reasons for dim results (contact vs resistance).
3. Label a real object (cord/plug) showing conductor and insulator parts with one safety reason.

Next Steps -- Where will this lesson lead to next?

1. Move into specific expectation D2.3 by building simple DC circuits and identifying component functions, using today's results to justify why wires must be conductors and why the coatings are insulators.
2. Use exit tickets to group students for a quick re-teach/troubleshooting station (fair test, dim LED causes) before the next activity.